

stop and relay RL1 de-energise to switch the appliance off.

Construction and working

An actual-size, single-side PCB for the microcontroller-based time-controlled switch is shown in Fig. 2 and its component layout in Fig. 3.

Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use an IC base for microcontroller. Before inserting the IC, check the supply voltage.

The time-controlled switch works in two modes: setting mode (to set the time from 1 to 99 minutes) and working mode (to drive the load for the desired time as per the setting). The modes can be changed using switch S3. When LED1 glows, it indicates that the system is in working mode. If LED1 is off, the system is in setting mode.

Setting mode. By default, when the microcontroller is powered up, it is in setting mode. In this mode, one of the two 7-segment displays should blink with a random digit, say, 6. You can change the blinking digit to any digit from 0 through 9 using decrement switch S4 or increment switch S5. You can also shift the blinking digit from DIS1 to DIS2 or vice-versa using switch S3. Thus the desired time can be set using switches S3, S4 and S5.

After setting the desired time, press start/stop switch S2 to switch from setting mode to working mode. The

appliance will turn on for the preset time unless you press switch S2 again to stop the operation in between.

Working mode. When the microcontroller is powered up, and neither of DIS1 and DIS2 blinks, the system is in working mode. In working mode, the digits on display show the time left for which the appliance will be 'on.' During this time, the microcontroller generates the actuating signal to energise relay RL1. LED1 acts as the load-'on' indicator and resistor R18 limits the current flowing through LED1.

In working mode, when you press switch S3 at any time, the display shows the set time for a second. When

the set time is over, relay RL1 de-energises to turn the appliance off and the mode changes to setting mode automatically.

If you want to switch the appliance 'off' during working mode, press start/stop switch S2. Working mode will change to setting mode and the relay de-energise to turn the appliance 'off.'

Software

The source program is written in Assembly language using instruction set mnemonics of PIC microcontroller and assembled using MPLAB integrated development environment (IDE). Burn the generated hex code from MPLAB IDE into the PIC16F72 chip using a standard programmer such as PIC-START Plus programmer from Microchip. The entire code is well commented and easy to understand.

The program execution starts by clearing the registers. The input and output ports are initialised by enabling the interrupt pin and global interrupts. This initialisation allows execution of the interrupt service routine when an interrupt is received. Interrupt subroutine is followed by B2BCD (binary to BCD), display, second, blinking_seg and minute subroutines. The code ends with the switching_delay subroutine.

EFY note. The source code of this article is available on www.efymag.com website. ●

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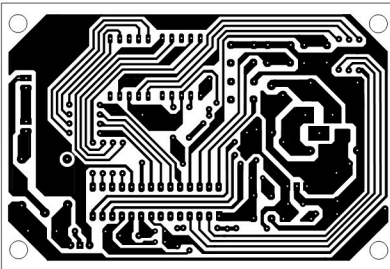


Fig. 2: An actual-size, single-side PCB for the time-controlled switch using PIC16F72

